

Project Topic: Artificial Intelligence In Education

Sr. No.	Paper Title	Journal Name
1	Adaptive AI Agent Systems for Personalized Learning: Frameworks, Algorithms, and Practical Applications in Education	Proceedings of International Conference on Artificial Intelligence and Education (ICAIE)
2	Agentic Artificial Intelligence in Education: A Review on Personalized Adaptive Learning Through Autonomous Tutoring Systems	International Journal of Applied Resilience and Sustainability
3	Artificial Intelligence in Education: Trends, Benefits, and Challenges (Systematic Review)	Multimodal Technologies and Interaction
4	The Transformative Impact of Artificial Intelligence on Higher Education: A Critical Reflection	Education Sciences

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Literature Review Summary

Publication House	Publication Year (2025-2026)	Dataset Details (Name /Sample size / weblink etc)
IEEE (Institute of Electrical and Electronics Engineers)	2025	<u>Dataset Type:</u> <u>Experimental + Case Study</u> https://ieeexplore.ieee.org/abstract/document/11158641 <u>Sample Size: 30 students (ESL course)</u>
Deep Science Publisher	2026	<u>Dataset Type:</u> Literature Review (Secondary Data) <u>Methodology:</u> PRISMA-based systematic review <u>Data Sources:</u> IEEE Xplore, ACM Digital Library, Springer, ScienceDirect, JSTOR, Google Scholar http://deepscipub.com/ijars/article/view/50
MDPI	2025	Dataset Type: Systematic Literature Review Total Papers Identified: ~3406 studies Final Papers Selected: 155 research papers Databases Used: Web of Science, Scopus, Taylor & Francis https://www.mdpi.com/2414-4088/9/8/84
MDPI	2025	Dataset Type: Conceptual / Critical Review Nature: Secondary data Sources Used: Existing research papers, policy reports, and case studies on AI in higher education https://journals.sagepub.com/doi/full/10.1177/2212585X251319364

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Algorithms / Models / Techniques implemented	Results Analysis (Various performance parameters)
Multimodal AI (text + voice + behavioral data) AI Agent Systems (Agentic learning framework) Machine Learning (for personalization & prediction) Natural Language Processing (NLP) (chatbots, language learning) Speech Recognition (pronunciation analysis) Adaptive Learning Algorithms Automated Design of Agentic Systems (ADAS) Gamification techniques (badges, rewards,	Vocabulary retention improved: 68% → 85% Pronunciation accuracy improved: 70% → 88% Speaking fluency increased: 100 → 135 WPM Engagement doubled: 45 → 80 mins/week Student satisfaction increased: 70% → 94% Overall: AI system significantly outperformed
Reinforcement Learning (Deep Q-Networks, Policy Gradient, Actor-Critic) Machine Learning & Deep Learning Natural Language Processing (NLP) Knowledge Tracing Algorithms Affective Computing (emotion detection) Cognitive Modeling (student behavior modeling) Multi-Agent Systems (hierarchical tutoring agents)	Significant improvement in personalized learning outcomes Better learning efficiency and knowledge retention Enhanced student engagement and motivation Effective real-time knowledge tracking and adaptive feedback Improved procedural skill development
Generative AI (most widely used) Machine Learning & Deep Learning Conversational AI / NLP Agents (chatbots) Intelligent Tutoring Systems (ITS) Rule-based & Expert Systems Methodology: PRISMA framework Immersive Systems (AR/VR, simulations)	Learning gains improved in 65.8% of studies Personalized learning reported in 26.5% of studies Improved problem-solving and critical thinking Increased student engagement and motivation Enhanced knowledge retention and digital literacy
Artificial Intelligence (general applications in education) Machine Learning (ML) Natural Language Processing (NLP) Generative AI (e.g., content creation, academic assistance) Learning Analytics systems Intelligent Tutoring Systems (ITS)	Increased efficiency in academic and administrative processes Improved accessibility of education through digital platforms Enhanced student support systems (feedback, tutoring, advising) Greater scalability of learning resources Mixed impact on academic integrity (rise in AI misuse concerns)

Observations (min 4-5 Bulleted points)	Research Gaps (min 4-5 Bulleted points)
AI agents provide highly personalized learning paths for each student Multimodal AI improves understanding by combining text, speech, and behavior data Real-time feedback enhances learning speed and accuracy Gamification increases student motivation and engagement	Small sample size (only 30 students → limited generalization) Limited testing across different subjects beyond ESL/STEM Ethical concerns: data privacy and algorithm bias not deeply addressed High dependency on advanced infrastructure and computational resources
Agentic AI systems act as autonomous tutors, not just tools Multi-agent systems improve scalability and complexity handling AI can adapt teaching based on cognitive + emotional state of students Personalized learning is more effective than traditional uniform teaching	Lack of algorithm transparency (black-box problem) High data requirements and cold-start problem Issues with privacy, bias, and fairness in AI systems Limited research on long-term learning impact and transfer learning
Generative AI is the most dominant technology in modern education AI significantly improves learning outcomes and engagement Personalized learning is a key benefit across most studies Language learning is the most researched domain	Underrepresentation in social sciences, humanities, and creative fields Lack of research on teacher training and AI integration in pedagogy Ethical concerns: bias, privacy, and data security High implementation cost and technical complexity
AI is reshaping higher education by making it more flexible and accessible Generative AI tools are changing how students learn, write, and research Institutions are increasingly relying on AI for decision-making and automation AI creates opportunities for personalized and self-paced learning There is a growing concern about over-reliance on AI tools by students	Lack of clear policies and regulations for AI use in education Limited research on long-term impact of AI on critical thinking skills Concerns about academic integrity and plagiarism using generative AI Ethical issues like bias, fairness, and data privacy not fully addressed

